Project Report

Final Project Report

Team Enrollment Confirmation – Import Data using Transform Maps (Spreadsheet)

Team Leader	SRIKAVI T
Team Member	ISWARYA S
Team Member	YOGASRI S
Team Member	KEERTHIKA L
Team Size	4

1. INTRODUCTION

1.1 Project Overview

Team Enrollment Confirmation is a platform application that imports team enrollment data from a spreadsheet using Import Sets and Transform Maps. Instead of typing each enrollment manually, the coordinator uploads an Excel/CSV file; the data is staged, cleaned, mapped to the Team Enrollment table, and each valid enrollment is marked as confirmed with a notification sent to the member.

1.2 Purpose

- Reduce manual data entry and errors during team enrollment.
- Provide a repeatable, auditable way to bulk-load and confirm enrollments.
- Avoid duplicate enrollments through coalesce rules.
- Give coordinators a quick view of enrollment status through lists and reports.

2. IDEATION PHASE

2.1 Problem Statement

Coordinators receive team enrollment details in spreadsheets and re-type them into the system one by one. This is slow, causes typing mistakes and duplicate entries, and delays confirmation to the team members.

Problem Statement	I am	I am trying to	But	Because	Which makes me feel
PS-1	A team coordinator	Enroll many teams and members quickly	Entries are typed manually and get duplicated or wrong	There is no bulk import and validation	Frustrated and unsure the data is correct
PS-2	A team member	Get confirmation of my enrollment	I hear back late or not at all	Confirmation is sent manually	Anxious and uninformed

2.2 Empathy Map Canvas

Quadrant	Coordinator / Team Member
Says	“I have the list in Excel, why do I need to type it again?” “Has my team been confirmed?”
Thinks	Manual entry will lead to mistakes; wants a fast and reliable way to upload data.
Does	Copies rows from a spreadsheet, checks for duplicates by eye, sends confirmations by email.
Feels	Overloaded near deadlines, worried about missing or duplicate records.
Pains	Repetitive work, errors, delayed confirmation, no single status view.

Quadrant	Coordinator / Team Member
Gains	One-step upload, clean data, automatic confirmation, clear reports.

2.3 Brainstorming

The team (Srikavi T, Iswarya S, Yogasri S, Keerthika L) brainstormed ideas and shortlisted the following:

Idea	Proposed by	Outcome
Bulk import from Excel using Import Set + Transform Map	Team	Selected
Coalesce on Team ID + Member Email to avoid duplicates	Team	Selected
Auto-set status and confirmation date in transform script	Team	Selected
Confirmation email through event notification	Team	Selected
Enrollment status reports / dashboard	Team	Selected
Self-service portal for members	Team	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Coordinator action	Touchpoint	Experience / Opportunity
Collect	Receives enrollment sheet	Spreadsheet	Standard template avoids format problems
Upload	Loads file via Load Data	Data Source / Import Set	One-step upload replaces re-typing
Transform	Runs Transform Map	Transform Map	Validation and duplicate control
Review	Checks results and errors	Team Enrollment list	Quick status view
Confirm	Enrollment confirmed automatically	Notification	Members informed instantly

3.2 Solution Requirement

Functional requirements

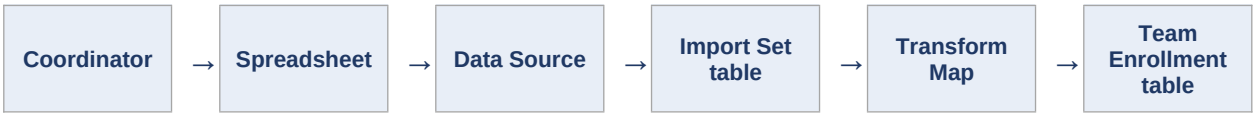
ID	Requirement	Description
FR-1	Upload spreadsheet	Load .xlsx/.csv through a Data Source into an Import Set table
FR-2	Field mapping	Map spreadsheet columns to Team Enrollment fields via Transform Map
FR-3	Duplicate control	Coalesce on Team ID and Member Email; update instead of insert

ID	Requirement	Description
FR-4	Confirmation	Set Status = Confirmed and Confirmation Date automatically
FR-5	Notification	Send confirmation email to the member / team leader
FR-6	Reporting	List and report of enrollments by status and team

Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	Simple upload and transform steps for non-technical coordinators
NFR-2	Security	Role-based access; only authorised users can import and view data
NFR-3	Reliability	Errors logged per row; valid rows still processed
NFR-4	Performance	Files of a few thousand rows transform without timeouts
NFR-5	Maintainability	Configuration-based maps and scripts that are easy to change

3.3 Data Flow Diagram



Level 1 flow: the coordinator uploads the spreadsheet; the Data Source loads rows into the staging Import Set table; the Transform Map validates, cleans and maps each row (coalescing on Team ID + Member Email); results are inserted or updated in the Team Enrollment table; an event triggers the confirmation notification; reports read from the target table.

3.4 Technology Stack

Layer	Technology
Platform	ServiceNow (Personal Developer Instance)
Data input	Excel (.xlsx) / CSV spreadsheet
Import	Data Source, Import Set, Transform Map
Logic	JavaScript transform scripts, business rules, events
Storage	Platform tables (staging and target)
Interface	Standard UI forms, lists, reports; REST APIs
Communication	Email notifications

4. PROJECT DESIGN

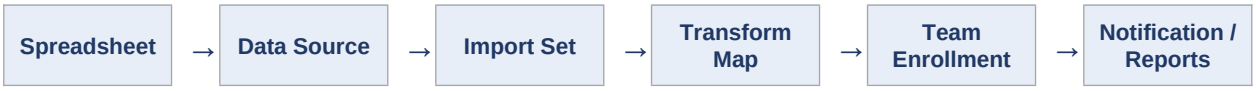
4.1 Problem Solution Fit

Problem	Solution fit
Manual re-typing of enrollments	Bulk import with a Transform Map
Duplicate and inconsistent records	Coalesce fields and cleaning scripts
Delayed confirmation	Automatic status update and notification
No status overview	Team Enrollment list and reports

4.2 Proposed Solution

Item	Description
Idea / Solution	Import team enrollment data from a spreadsheet using Transform Maps and auto-confirm it
Novelty	Configuration-based import with coalesce and confirmation in one flow
Social impact	Saves coordinator time and gives members faster, reliable confirmation
Business model	Reusable internal tool for colleges, clubs and events
Scalability	Same Transform Map handles small or large files; can be scheduled

4.3 Solution Architecture



The architecture has three layers: input (spreadsheet and data source), processing (import set staging and transform map with scripts), and output (target table, notifications and reports), secured by roles and ACLs.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Task	Owner	Status
Sprint 1	Requirement analysis, problem statement, empathy map	SRIKAVI T	Completed
Sprint 1	Create Team Enrollment table, form and list layout	YOGASRI S	Completed
Sprint 2	Prepare spreadsheet template; create Data Source and Import Set	ISWARYA S	Completed
Sprint 2	Create Transform Map, field maps and coalesce rules	SRIKAVI T	Completed

Sprint	Task	Owner	Status
Sprint 3	Transform scripts, notification, ACLs	YOGASRI S	Completed
Sprint 3	Testing, UAT and defect fixing	KEERTHIKA L	Completed
Sprint 4	Reports, documentation and final report	KEERTHIKA L, ISWARYA S	Completed

Adjust sprint dates, owners and status to match your actual schedule.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test scenario	Input	Expected result	Result
Small file import	10 rows	All rows inserted and confirmed	Pass
Duplicate import	Same file loaded twice	Records updated, no duplicates	Pass
Invalid data	Blank email / wrong date	Row flagged in error list; others processed	Pass
Larger file	1,000 rows	Transform completes without timeout	Pass
Notification	Confirmed record	Email event fired to member	Pass
Access control	User without role	Import and data access denied	Pass

Sample results shown; replace with your actual test observations before submission.

7. RESULTS

7.1 Output Screenshots

[Insert screenshot: Data Source and uploaded spreadsheet]

[Insert screenshot: Transform Map field mappings]

[Insert screenshot: Import Set rows after transform]

[Insert screenshot: Team Enrollment table with confirmed records]

[Insert screenshot: Confirmation notification / email]

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Saves time by importing records in bulk	Spreadsheet format must follow the template
Reduces typing errors and duplicates	Large files need to be split into batches
Automatic confirmation and notification	Error correction is done manually in the source file
Configuration-based and easy to modify	Requires an admin to create and maintain the Transform Map
Audit trail through import set history	Depends on platform instance availability

9. CONCLUSION

The project shows that Import Sets and Transform Maps can replace manual entry of team enrollments. Spreadsheet data is loaded, cleaned, checked for duplicates, mapped to the Team Enrollment table and confirmed automatically, which saves time, improves accuracy and gives members faster confirmation.

10. FUTURE SCOPE

- Scheduled imports from a shared location or record attachment.
- Error summary email after each import.
- Self-service portal for members to view, confirm or decline enrollment.
- Dashboards for enrollment trends and team participation.
- Integration with approvals and external systems.

11. APPENDIX

Source Code (if any)	Transform Map scripts (onBefore/onAfter) and update set: [add file / repository link]
Dataset Link	Sample enrollment spreadsheet: [add link]
GitHub & Project Demo Link	GitHub: [add link] Demo video: [add link]